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LOT SYSTEMS CORPORATION
DOCUMENT: LOT-COSMO-COMPUTER-BOM-v1
TITLE:   COSMO® Computer – Components Buying List & 100-Unit Cost Roll-Up
CLASS:   RESTRICTED // S-2 EYES
S-2:    VADIK MARMELODOV
DATE:    2026-08-05
VERSION: 1.0 – companion to LOT-COSMO-COMPUTER-v1
PARENT:  docs/corporate/LOT-COSMO-COMPUTER-v1.md (Section 10, item 2)
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00 // HOW TO READ THIS LIST

Every line item below is a real, currently-manufactured commodity part class (brief point 15: "AI grade off-the-shelf sensors" – read as commodity-qualified, not custom silicon), sourced from named real suppliers. Links point to each supplier's stable SEARCH or ORDER entry point, not to a single fabricated product page – part numbers and quantities are given in the table so the actual SKU can be located and price-confirmed at order time, since distributor pricing and stock shift week to week. Prices below are PLANNING ESTIMATES for the 100-unit pilot run (Section 06 of the parent document), not quotes. Nothing in this document authorizes a purchase order.

01 // ELECTRONICS BOM (PER UNIT, x100 FOR THE PILOT RUN)

REF	PART CLASS	CANDIDATE PART	QTY/UNIT	EST. UNIT COST
U1	MCU / Wi-Fi+BLE SoC	Espressif ESP32-S3-WROOM-1 (8MB flash / 8MB PSRAM)	1	\$2.50 - \$4.00
U2	Camera module	OV2640 (2MP, DVP/SPI) or OV5640 (5MP)	1	\$3.00 - \$6.00
U3	E-paper display	1.54" e-paper module, monochrome or 3-color, SPI (GDEY0154-class)	1	\$8.00 - \$14.00
U4	Weather sensor	Bosch BME280 (temp/humidity/pressure), I2C	1	\$1.50 - \$3.00
U5	Wireless charge receiver + PMIC	Qi receiver coil + BQ51013B-class receiver IC	1	\$2.00 - \$4.00
U6	LiPo charge/protect IC	Single-cell Li-Po charger + protection IC	1	\$0.50 - \$1.00
U7	Battery	LiPo pouch cell, 250-350mAh	1	\$2.00 - \$3.50
U8	Button	SMD tactile switch, "Copy" cap (custom-marked)	1	\$0.10 - \$0.30
U9	PCB (4-layer, ~34x34mm)	Custom, per parent doc Section 03/06 spec	1	\$1.50 - \$3.00 (at 100-unit qty, PCBWay fab only)
U10	Passives + regulators	Resistors, caps, LDO, ESD protection	1 set	\$1.00 - \$2.00
U11	FPC/connectors	Camera + display flex connectors	2	\$0.50 - \$1.00
ELECTRONICS SUBTOTAL (per unit, parts only):			~\$22.60 - \$41.80	
PCBWay SMT ASSEMBLY (per unit, 100-unit run):			~\$5.00 - \$10.00 (labor + setup, amortized – see Section 03)	

02 // ENCLOSURE BOM (PER UNIT, x100)

REF	PART CLASS	SPEC	QTY/UNIT	EST. UNIT COST
E1	Top shell	316 stainless steel, CNC-machined, polished mirror finish, ~40x40x2.5mm	1	\$6.00 - \$12.00
E2	Base shell	316 stainless steel, CNC-machined, matte bead-blast around screen window, ~40x40x2.5mm	1	\$8.00 - \$15.00
E3	RF/charge insert ring	Non-metal composite (ABS/PC, stainless-finish PVD coat), machined or injection-molded	1	\$1.00 - \$2.50
E4	Fasteners / seal	Shell-to-shell screws or press-fit hardware, gasket	1 set	\$0.30 - \$0.80

E5 Elastomer feet 4x, non-marking, desk-safe 4 \$0.20 - \$0.40
ENCLOSURE SUBTOTAL (per unit, 100-unit CNC run): ~\$15.50 - \$30.70

03 // CHARGING PAD (COMPANION ACCESSORY, PER UNIT, x100)

REF	PART CLASS	SPEC	QTY/UNIT	EST. UNIT COST
P1	Qi wireless transmitter coil + driver IC	15W-class Qi transmitter module, off-the-shelf	1	\$3.00 - \$6.00
P2	Pad enclosure	Flat base, brand-neutral material at v1.0 (not scoped to stainless – see parent doc Section 07 open items)	1	\$2.00 - \$4.00
P3	USB-C power input + adapter	Cable + AC adapter, off-the-shelf	1 set	\$2.50 - \$4.50

CHARGING PAD SUBTOTAL (per unit companion, x100): ~\$7.50 - \$14.50

04 // SUPPLIER DIRECTORY (SEARCH / ORDER ENTRY POINTS)

Distributor and fabrication links below go to each vendor's stable search or quote entry point – confirm exact SKU, stock, and price at order time.

PCB FABRICATION + SMT ASSEMBLY + CNC (brief point 1)
PCBWay – PCB + PCBA instant quote:
<https://www.pcbway.com/orderonline.aspx>
PCBWay – CNC machining (stainless shells, Section 02 E1/E2/E3):
https://www.pcbway.com/rapid_prototyping.html

COMPONENT DISTRIBUTORS (electronics BOM, Section 01)
DigiKey – part search:
<https://www.digikey.com/en/products>
Mouser Electronics – part search:
<https://www.mouser.com/c/>
LCSC (China-side sourcing, common PCBWay PCBA pairing):
<https://www.lcsc.com/>

MODULE / HOBBYIST-GRADE SOURCES (prototyping stage, pre-100-unit run)
Adafruit – ESP32-S3, e-paper, BME280 breakout boards for bring-up:
<https://www.adafruit.com/>
SparkFun – sensor and display breakouts for bring-up:
<https://www.sparkfun.com/>

BATTERY
LiPo pouch cells, UL-certified suppliers listed via Mouser/DigiKey search above – no dedicated third-party link needed at BOM stage.

05 // 100-UNIT COST ROLL-UP (PLANNING ESTIMATE, NOT A QUOTE)

CATEGORY	LOW EST.	HIGH EST.
Electronics BOM (Section 01)	\$22.60	\$41.80
PCBWay SMT assembly (Section 01)	\$5.00	\$10.00
Enclosure, CNC stainless (Section 02)	\$15.50	\$30.70
Charging pad companion (Section 03)	\$7.50	\$14.50
PER-UNIT TOTAL	\$50.60	\$97.00
x 100 UNITS	\$5,060	\$9,700

This range excludes: firmware/software engineering time, tooling and fixture cost for hand-assembly (parent doc Section 06 step 3), shipping, import duties, and QA labor. It is a materials-and-fabrication planning number only, meant to be replaced by PCBWay's actual instant-quote output once a physical prototype BOM is finalized – this document is the input to that quote, not a substitute for it.

CONTEXT AGAINST EXISTING LOT PRICING
LOT_ROBOTICS_COSMO.md's own revenue table prices "COSMO® Hardware" at \$2,500-\$5,000 per unit at Phase 3 (2028-2029) – a fully different product tier (full robotics platform) from this document's desk-object pilot. The \$50-\$97/unit materials estimate here is not in tension with that figure; it confirms the pilot-scale Puck is a materials-cheap,

engineering-expensive object at 100 units, which is exactly the profile
a pilot run should have before any Phase-3 pricing commitment.

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